

FCPS dissertation research protocol																			
Protocol No																			
1	Date of submission of protocol																		
2	Relevant faculty of BCPS: Surgery	Relevant Discipline								Anaesthesiology									
3	Name of Examinee	D	R	.		P	A	R	T	H	A		P	R	A	T	I	M	
			S	E	N														
4	Address of the correspondence of the examinee & contact phone Number	Dr. Partha Pratim Sen Dept. of Anaesthesiology & Surgical ICU BIRDEM General Hospital & IMC, Shahbagh, Dhaka dr.parthapratimsen@gmail.com BMDC Reg. A-79867 Reg. No. of BCPS: 2019070309 Online Reg. No: A-2018-1-11-0046 Cell#01717447187, 01521466691																	
5	Title of the dissertation	COMPARISON OF GLYCEMIC STATUS WITHIN BI-SPECTRAL INDEX 40-60 INDUCED BY ISOFLURANE VS SEVOFLURANE ANAESTHESIA IN DIABETIC PATIENTS UNDERGOING MAJOR ABDOMINAL SURGERY																	
6	Summary	Diabetes is one of the important health issues all over the world. Higher prevalence rate of diabetes in Bangladesh makes it a bigger challenge. Surgery of a diabetic patient due to different cause is a potential source of morbidity and mortality as uncontrolled diabetes leads to end organ damage like kidney, heart, liver etc. The effect of surgical stress on diabetic patients increases the risks of adverse outcome. Perioperative hyperglycemia is one of the indicator of stress response. And type of anaesthesia has an impact on perioperative																	

		glycemic control. So, our aim is to find out appropriate anaesthetic agent which provides better glycemic control. It will be a prospective, single blind clinical trial that includes 56 diabetic patients. Patients will be divided into two groups, named S=Sevoflurane and I=Isoflurane. Anaesthesia protocol will be same for both groups except inhalational agent maintaining adequate anaesthetic depth. One group will receive sevoflurane and another isoflurane. Blood glucose levels and other clinical outcomes such as heart rate, blood pressure, end tidal CO₂, bi-spectral index etc. will be measured at different interval. Blood glucose level will be measured 3 times perioperatively such as pre-induction period, 30 minutes after incision and immediately after recovery by Simens AVIDA 1800 Chemistry System. Bi-spectral index value will be maintained within the value of 40-60 and documented at 10 minutes' interval along with concentration of inhalational agent. The measured data will be analyzed and used to compare the two groups. At the end of the study, we expect to obtain a comparison between isoflurane and sevoflurane anaesthesia on the glycemic status.
7	Study setting/ place of study	Department of Anaesthesiology and Surgical ICU, BIRDEM General Hospital, Shahbagh, Dhaka-1000
8	Study period	One year
9	Overview of the study design	Prospective, single blind clinical trial
10	Introduction /context	Diabetes is one of the common non communicable

	diseases around the globe. In 2017, according to International Diabetes Federation, more than 425 million people were affected from diabetes (1). Prevalence of diabetes in adults was 8.8% worldwide in 2015 and it may rise to 10.4% in 2040 (2). But in Bangladesh scenario, prevalence of diabetes is already 11% in adult men and 10% in adult women in 2017 (3). And it is gradually increasing from previous studies. Many socio-economic and clinical factors are working behind it. Moreover, proportion of adults having undiagnosed diabetes is about 52.1% in south-east Asia which is second highest in the world (2). Type 2 diabetes constitutes around 90% of all diabetic cases (1). In 2015, global expenses for management of diabetes was 673 billion USD (2). This burden increases when a diabetic patient undergoes surgery. Diabetes is present in 21% of surgical population and it rises to 40% in patients aged above 65 years (4). In some recent studies, peroperative control of blood glucose level is one of the major factors for reducing complications and length of hospital stay and thereby, extent of expenses (5). Surgical stress results in cytokine and neuroendocrine response(6). These responses activate counter regulatory hormones such as catecholamines, cortisol, glucagon etc. There is also decreased level of insulin in blood due to these responses as well as insulin resistance. Blood glucose level is related to extent of surgical injury and laryngoscopic stress (6). So, final result appears as alteration of glycemic status. Diabetic patients are more prone to peroperative hyperglycemia because they already lack insulin or having insulin resistance to some degree. In a study in 55 hospitals of United States, incidence rate of peroperative
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	hyperglycemia is 40% in diabetic patients (7). There is increasing evidence that use of different anaesthetic agent has an impact on blood glucose level (8)(9)(10). Moreover, moderate glycemic control target (8.3-11.1 milimole/liter) is associated better outcome in terms of mortality and morbidity (11). There are very few studies regarding peroperative glycemic status in Bangladesh. Glycemic response under halothane anaesthesia was investigated in one study (12). Studies related to comparison between isoflurane and sevoflurane: and their impact on peroperative glycemic status are even lower in number. Moreover, in past studies there is limited use of anaesthetic depth measuring system such as bi-spectral index. This bi-spectral index is a processed EEG obtained from frontal electrode to provide a measure of anaesthetic depth and it produces dimensionless number, ranging from 0-100. As we know inadequate anaesthetic depth is also a source of stress response, so incidence rate of peroperative hyperglycemia from previous studies needs to be evaluated in current scenario. In a recent study in USA, mean blood glucose level before the surgery was 146.4 ± 51.9 mg/dL and after the surgery 139.9 ± 45.6 mg/dL but anaesthetic depth is not monitored in the study (5). Moreover, peroperative insulin and glucose containing fluid administration are confounding factors for blood glucose level reduction along with anaesthetic depth in the study (5). So, our plan is to maintain adequate anaesthetic depth to reduce or obtund stress response, which consequently leads us to reduce blood sugar and getting a better glycemic control without administration of any hypoglycemic agent and glucose containing fluid peroperatively.
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		We choose to observe glycemic status in patients who require elective major abdominal surgery. Risk of perioperative hyperglycemia is higher in major surgeries as they have more surgical stress. Institutional availability is the reason of choosing abdominal surgeries such as triple bypass, whipple's procedure, hemicolectomy, total colectomy, anterior resection, abdominal-perineal resection, partial gastrectomy etc. We will select the patients who are treated with insulin therapy to control diabetes preoperatively. We plan to provide same anaesthetic and recovery protocol to every patient except inhalational agent. Every patient will receive induction with propofol (2mg/kg), fentanyl (2µgm/kg), atracurium (0.5mg/kg). Maintenance of muscle relaxation will be achieved by incremental dose of atracurium (0.1mg/kg) at every 20-minute interval. Depth of anaesthesia will be ensured by inhalational agent (Isoflurane or Sevoflurane) with the dose of 1 MAC (Minimum Alveolar Concentration) but dose will be adjusted to maintain bi-spectral index value within 40-60. Intraoperative pain will be managed by using nitrous oxide as carrier gas. Nitrous oxide will be maintained along with oxygen at a ratio of 2:1 (N₂O:O₂). Normal saline (0.9% NaCl) will be used as maintenance fluid perioperatively. Recovery protocol for reversing muscle relaxation is to administer neostigmine (0.04 mg/kg) after observing any muscular activity such as starting of spontaneous breathing effort, eye opening, tongue movement, grimace, gag reflex, hand gripping etc. But dose of neostigmine will be targeted to achieve adequate spontaneous tidal volume (5-10ml/kg). Anti-secretory drug such as atropine (0.02mg/kg) will be required to prevent excessive salivary secretion and bradycardia due to neostigmine administration. We will
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		not use any regional anaesthetic technique such as epidural block or sub-arachnoid block in the study. Our objective is to study the peroperative glycemic status under adequate anaesthetic depth, compare isoflurane and sevoflurane groups about their impact on glycemic status and observe any complications between these groups.
11	Rationale of the research	Peroperative control of blood glucose level is one of the major factors for reducing complications and length of hospital stay and thereby, extent of expenses. Blood glucose level is related to extent of surgical injury and laryngoscopic stress peroperatively. Surgical and laryngoscopic stress results in cytokine and neuroendocrine response. These responses activate counter regulatory hormones such as catecholamines, cortisol, glucagon etc. There is also decreased level of insulin in blood due to these responses as well as insulin resistance. So, final result appears as peroperative fluctuation of blood glucose level. In modern anaesthesiology, intravenous and inhalational agent is used to reduce surgical and laryngoscopic stress by achieving depth of anaesthesia. In previous studies, relation of blood glucose level and surgical stress was observed by using different anaesthetic agent. But depth of anaesthesia was not measured due to lack of high-tech facilities such as bi-spectral index monitor. So, now we intend to observe glycemic status of diabetic patients who are undergoing major abdominal surgery with the measurement of anaesthetic depth by bi-spectral index monitor. And here we use isoflurane and sevoflurane as inhalational agent and compare their effectiveness regarding controlling blood glucose level.

12	Research question / hypothesis	Peroperative glycemic status is not equal in isoflurane and sevoflurane anaesthesia	
13	Objectives	General objective: To observe the impact of isoflurane and sevoflurane anaesthesia on peroperative glycemic status of diabetic patients Specific objectives: 1. To observe fluctuation of peroperative blood glucose level 2. To compare fluctuation of peroperative glucose level under isoflurane and sevoflurane anaesthesia 3. To manage diabetic patient more safely 4. To assess the complications between two groups	
14	Material & Method	a. Main outcome variable to be studied	Peroperative blood glucose level
		b. Confounding variables, if applicable	Faulty technique, inadequate pain relief, intraoperative awareness, drug metabolism, temperature, shivering
		c. Study population (mentioning species & strains, ethnic group, age, sex & other relevant character)	Diabetic patients of BIRDEM General Hospital requiring major abdominal surgery. They may be of different sexes and ages but above 18 years.
		d. Sample size & the statistical basis of it	Sample size, N = $\frac{p_s(100-p_s) + p_i(100-p_i)}{(p_s-p_i)^2} \times (Z_{\alpha} +$

			$Z_{\beta})^2$ P_s = incidence rate of peroperative hyperglycemia under sevoflurane anaesthesia P_i = incidence rate of peroperative hyperglycemia under isoflurane anaesthesia Z_{α} = z-value of standard normal distribution at 95% confidence level Z_{β} = z-value of standard normal distribution at 95% power Here, $P_s = 72\%$ $P_i = 40\%$ $Z_{\alpha} = 1.96$ $Z_{\beta} = 0.84$ $N =$ $\frac{72(100-72)+40(100-40)}{(72-40)^2}$ $\times (1.96+1.64)^2$ $= \frac{72 \times 28 + 40 \times 60}{32^2} \times (3.6)^2$ $= \frac{2016+2400}{1024} \times 12.96$
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			$= 4.3125 \times 12.96$ $= 55.89$ N=56
		e. Screening method, if applicable	After doing all necessary investigations, 56 patients will be selected in the pre-anaesthetic checkup room who are suffering from diabetes mellitus and going to be operated. Informed written consent will be taken from all selected patients. The anaesthetic procedure will be explained the day before the operation to all patients.
		f. Sampling method (s)	Purposive Sampling
		g. Inclusion & exclusion criteria	The inclusion criteria included: (1) Diabetic patients who require elective major abdominal surgery under general anaesthesia (2) Diabetic patients under insulin therapy (3) Those who are belonging to ASA II and ASA III physical status (4) The patient who are adult (age above 18 years)

			The exclusion criteria included: (1) Patient's refusal (2) Emergency or septic abdomen (3) Presence of a Psychiatric history (4) Acute or chronic renal failure requiring dialysis (5) Altered liver function (6) Preoperative fasting blood glucose more than 10mmol/liter on operating day (7) Preoperative fasting blood glucose less than 5mmol/liter on operating day (8) Medication known to affect blood glucose and insulin release, e.g. Steroids, Beta Blockers
		h. Operational definition	Major Abdominal Surgery: Any operation within or upon the contents of abdominal cavity which may include removal or resection with or without reconstruction with or without anastomosis with or without diversion of abdominal viscera and constitutes hazard to life or function of organ or tissue.
		i. Flow chart showing the	Target population

		sequence of task (After willingly given the informed written consent, adult diabetic patients requiring major abdominal surgery of ASA Class- II, III of BIRDEM General Hospital) ↓ Enrollment of the patient by inclusion & exclusion criteria ↓ Divide into group S and group I ↓ Administration of Anaesthesia ↓ Evaluation of Blood Glucose at different interval such as pre-induction period, 30 minutes after incision and immediately after recovery as well as bi-spectral index, anaesthetic gas concentration, blood pressure, heart rate, fingertip oxygen saturation, end tidal CO₂ will be evaluated every 10-minute interval ↓ Recording data in data collection sheet ↓ Data analysis & report writing ↓ Printing & binding of report
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			↓ Submission of report to BCPS
		j. Procedure of preparing & organizing material (e.g. : into groups)	Fifty-six (56) patients in the current study will constitute the study sample. They may be of different sexes and ages (above 18 years). These patients will be divided into two groups by purposive sampling. The first participant patient will choose a lottery token from two tokens (one for Group S and another for Group I). Then we will select the patient in alternating manner, e.g. One from Group S then next one from Group I. There will be twenty-eight (28) patients in each group. Group – S: Twenty-eight (28) patients will receive Sevoflurane Group – I: Twenty-eight (28) patients will receive Isoflurane
		k. Nature of controls, if applicable	Not applicable
		l. Randomization & binding methods, if & as	Randomization: The first participant patient will choose a lottery token from two tokens

		applicable	(one for Group S and another for Group I). Then we will select the patient in alternating manner, e.g. One from Group S then next one from Group I. There will be twenty-eight (28) patients in each group Single Blind Method: only the participants are kept unknown about the inhalational agent used for anaesthesia
		m. Equipment to be used	Blood glucose monitoring: 1. Simens AVIDA 1800 Chemistry System Vital sign and BIS monitoring: 1. Drager Vista 120 2. Drager scio four for anaesthetic gas analyzer Anaesthesia machine and vaporizer: 1. Neptune Anaesthesia workstation by Meditec England 2. Blease Datum Isoflurane Vaporizer by Spacelab technologies 3. Blease Datum Sevoflurane Vaporizer by Spacelab technologies Drugs: 1. Terrel (Isoflurane)-

			100ml; Manufactured by- Piramal (United States) 2. Sojourn (Sevoflurane)- 250ml; Manufactured by- Piramal (United States) 3. Pofol (Propofol); Manufactured by- Popular Pharmaceuticals (Bangladesh) 4. Opifen (Fentanyl); Manufactured by- Incepta Pharmaceuticals (Bangladesh) 5. Relaxton(Atracurium) Manufactured by- Techno Drugs (Bangladesh) 6. G-Neostigmine (Neostigmine); Manufactured by- Gonoshasthaya Pharmaceuticals (Bangladesh) 7. G-Atropine (Atropine) Manufactured by- Gonoshasthaya Pharmaceuticals (Bangladesh) 8. Normal Saline IV, Manufactured by- Opso Saline Ltd (Bangladesh)
		n. Procedure of collecting data	Data will be collected and compiled in a predesigned data

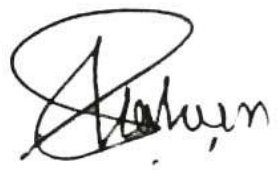
		(including methods of intervention, measurement, estimation etc.)	collection sheet.																	
		o. Professional assistance from expert, if applicable	Not applicable																	
		p. Procedure of data analysis of interpretation (including methods of statistical analysis – specific methods & software used for individual analysis)	All data will be collected in a pre-designed data collection form. Categorical data will be analyzed using Chi-square test. Continuous data will be analyzed using students ‘t’ test. P value <0.05 will be considered statistically significant.																	
		q. Quality assurance strategy	Quality assurance by meticulous monitoring by research guide and co-guide and self-presence																	
		r. Time table	<table><tr><th>Activity</th><th>Month-1st</th><th>Month 2nd - 7th</th><th>Month 8th - 10th</th><th>Month 11th -12th</th></tr><tr><td>Phase I 1.Literature Review 2.Development of questionnaire 3.Pretesting</td><td></td><td></td><td></td><td></td></tr><tr><td>Phase II 1.Data collection</td><td></td><td></td><td></td><td></td></tr></table>					Activity	Month-1st	Month 2 nd - 7 th	Month 8 th - 10 th	Month 11 th -12 th	Phase I 1.Literature Review 2.Development of questionnaire 3.Pretesting					Phase II 1.Data collection		
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Phase I 1.Literature Review 2.Development of questionnaire 3.Pretesting																				
Phase II 1.Data collection																				

			Phase III 1.Data entry 2.Edit 3.Data analysis				
			Phase IV 1.Report writing 2.Submission				
15	Ethical implication	The approval of the Institutional Review Board, BIRDEM General Hospital was duly taken before carrying out the study					
16	Total budget	Three (3) blood sample from each patient for blood glucose measurement: $3 \times 200 = 600/-$ Sample Size N= 56 Expense on blood glucose measurement= $56 \times 600 = 33,600/-$ Data sheet composition and printing, CD writing, dissertation book organizing and binding = 4000/- Informed written consent and other study materials composition and printing= 4000/- Personal protective equipment for patient and OT Staff= 9600/- Drugs: (Isoflurane, Sevoflurane, Atracurium, Propofol, Fentanyl, Atropine, Neostigmine) = 1,47,200/- Total: 1,98,400 TK (One lac ninety-eight thousand four					

		hundred taka only)
17	Source(s) of funding (if Any)	Self or if any source of funding is available (BCPS, BMRC)
18	Facilities available at the place of study	All facilities are available for study including sample, vital sign and bi-spectral Index monitoring, blood glucose monitoring, post-operative monitoring, emergency management including surgical ICU & medical ICU and biochemistry lab in BIRDEM General Hospital
19	Other facilities needed	Not applicable
20	Dissemination & use of the findings	Not applicable
21	Reference	1. International Diabetes Federation. IDF DIABETES ATLAS Eighth edition 2017 [Internet]. IDF Diabetes Atlas, 8th edition. 2017. 1–150 p. Available from: https://www.diabetesatlas.org/upload/resources/previous/files/8/IDF_DA_8e-EN-final.pdf 2. Ogurtsova K, da Rocha Fernandes JD, Huang Y, Linnenkamp U, Guariguata L, Cho NH, et al. IDF Diabetes Atlas: Global estimates for the prevalence of diabetes for 2015 and 2040. Diabetes Res Clin Pract [Internet]. 2017;128:40–50. Available from: http://dx.doi.org/10.1016/j.diabres.2017.03.024 3. National Institute of Population Research and Training (NIPORT) and I 2020. Bangladesh Demographic and Health Survey 2017-18 [Internet]. Dhaka, Bangladesh, and Rockville, Maryland, USA: NIPORT and ICF; 2020. Available from:

		https://dhsprogram.com/pubs/pdf/PR104/PR104.pdf 4. Cruz NI, Santiago E, Abdul-Hadi A. Prevalence of diabetes mellitus in the surgical population of the university of Puerto Rico affiliated hospitals: A study using the surgery database. P R Health Sci J. 2016;35(3):160–4. 5. Garg R, Schuman B, Bader A, Hurwitz S, Turchin A, Underwood P, et al. Effect of Preoperative Diabetes Management on Glycemic Control and Clinical Outcomes after Elective Surgery. Ann Surg. 2018;267(5):858–62. 6. ALAN R. AITKENHEAD, IAIN K. MOPPETT JPT. Smith and Aitkenhead's Textbook of Anaesthesia. 6th ed. Elsevier Limited; 2013. 7. Kotagal M, Symons RG, Hirsch IB, Umpierrez GE, Dellinger EP, Farrokhi ET, et al. Perioperative hyperglycemia and risk of adverse events among patients with and without diabetes. Ann Surg. 2015;261(1):97–103. 8. Akavipat P, Polsayom N, Pannak S, Punkla W. Blood glucose level in neurosurgery. Is it different between isoflurane and desflurane anesthesia? Acta Med Indones. 2009;41(3):121–5. 9. Behdad S, Mortazavizadeh A, Ayatollahi V, Khadiv Z, Khalilzadeh S. The effects of propofol and isoflurane on blood glucose during abdominal hysterectomy in diabetic patients. Diabetes Metab J. 2014;38(4):311–6. 10. Kim H, Han J, Jung SM, Park S-J, Kwon NK. Comparison of sevoflurane and propofol
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		anesthesia on the incidence of hyperglycemia in patients with type 2 diabetes undergoing lung surgery. Yeungnam Univ J Med. 2018;35(1):54–62. 11. Sathya B, Davis R, Taveira T, Whitlatch H, Wu WC. Intensity of peri-operative glycemic control and postoperative outcomes in patients with diabetes: A meta-analysis. Diabetes Res Clin Pract [Internet]. 2013;102(1):8–15. Available from: http://dx.doi.org/10.1016/j.diabres.2013.05.003 12. Sardar K, Ali L, Rashid MA, Faruque O. Hemodynamic , glycemic and electrolyte response to total abdominal hysterectomy under general anesthesia in type 2 diabetic subjects. DIABETES Endocr J [Internet]. 2008;(June). Available from: https://www.researchgate.net/publication/309827517_Hemodynamic_glycemic_and_electrolyte_response_to_total_abdominal_hysterectomy_under_general_anesthesia_in_type_2_diabetic_subjects
22	Any other relevant information	Not applicable
23	I solemnly pledge that this research protocol shall be implemented in accordance with the relevant ordinance / circulars of BCPS & funding agencies as & when it may be applicable. I hereby declare that no part of the proposed research has been used in any thesis / dissertation in partial fulfillment of any degree / fellowship or in any publication. I also understand that the BCPS reserves the right of accepting or rejecting this protocol.	

	 Date Signature	
24	Signature(s) of the supervisor(s)	Signature(s) of the Co supervisor
	 Dr. Kawsar Sarder MD(Anaesthesiology), FIPP Professor and Head of Dept. of Anaesthesia & Surgical ICU BIRDEM General Hospital & IMC Shahbagh, Dhaka Cell: 01819133800	 Dr. Shafiul Alam Shaheen (DA, FCPS) Assistant Professor Dept. of Anaesthesia & Surgical ICU BIRDEM General Hospital & IMC Shahbagh, Dhaka Cell: 01912078807